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1a) - Feature A

Known Values:

$F = 600 \text{ lbf}$, $P = 2F = 1200 \text{ lbf}$, $SF = 4$

Material: ASTM A36 Steel

$S_y = 36 \text{ ksi} = 36,000 \text{ psi}$

$$\sigma_{\text{allow}} = \frac{S_y}{SF} = \frac{36,000}{4} = \boxed{9,000 \text{ psi}}$$

1b) - Feature A

r_A = minimum required radius of Feature A

1c) - Feature A

1. Feature A is modeled as a cantilever beam, as specified in Appendix A

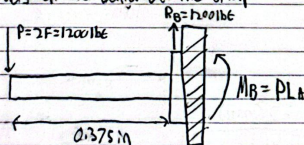
2. The load is static and applied symmetrically to Feature A

3. Direct shear stress is neglected, as stated in the assignment

4. ASTM A36 steel is assumed to be homogeneous, isotropic

5. Assume the 0.75 in wide strip is positioned directly against Feature B, so the resultant load acts at the center of the strip

1d)



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1e) - Feature A

$$\sigma = \frac{Mc}{I}, M = PL_A$$

Solid circular cross-section:

$$I = \frac{\pi r^4}{4}, c = r_A$$

$$\sigma_{allow} = \frac{(PL_A)(r_A)}{\frac{\pi r^4}{4}} \rightarrow \sigma_{allow} = \frac{4PL_A}{\pi r^3}$$

$$r_A = \sqrt[3]{\frac{4PL_A}{\pi \sigma_{allow}}}$$

1f) - Feature A

$$r_A = \sqrt[3]{\frac{4P_L}{\pi \sigma_{allow}}}$$

$$r_A = \sqrt[3]{\frac{4(1200)(0.375)}{\pi(4000)}}$$

$$r_A = \sqrt[3]{0.04366}$$

$$r_A = 0.40 \text{ in}$$

$$d_A = 2r_A \rightarrow d_A = 2(0.40)$$

$$d_A = 0.80 \text{ in}$$

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1a) - Feature B

Known values:

$$P = 2F = 1200 \text{ lbf}, SF = 4, S_y = 36,000 \text{ psi}$$

$$\sigma_{allow} = S_y / SF = 36,000 / 4 = 9,000 \text{ psi}$$

$$d_A = 0.80 \text{ in}, b_B = d_A = 0.80 \text{ in}$$

1b) - Feature B

t_B = minimum thickness of feature B

1c) - Feature B

1. Feature B is an axially loaded member

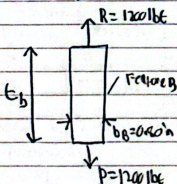
2. The 1200-lbf load acts through the centroid of B

3. Loading is static and symmetric

4. Direct shear stress is neglected

5. Feature B has a rectangular cross-section with: $b_B = 0.80 \text{ in}$

1d) - Feature B



1e)

$$\sigma = P/A, A = b_B t_B, \sigma_{allow} = \frac{P}{b_B t_B}$$

$$t_B = \frac{P}{b_B \sigma_{allow}}$$

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4(c) - Feature B

1200

$$t_B = \frac{1200}{(0.98)(9000)}$$

$$\rightarrow t_B = 0.17 \text{ in}$$

$$A_B = (0.98)(0.1667)$$

$$A_B = 0.133 \text{ in}^2$$

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1A) - Feature C

Known Values:

$P = 1200 \text{ lbf}$, $SF = 4$, $S_y = 36,000 \text{ psi}$

$$\sigma_{allow} = \frac{36,000}{4} = 9000 \text{ psi}$$

$W_c = 0.80 \text{ in}$

$L_c = 0.125$

$L_c = 0.499 + 2(0.9942) = 2.4864 \text{ in}$

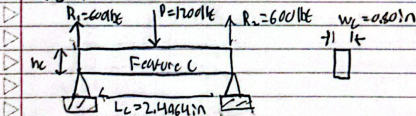
1B) - Feature C

h_c = Minimum required thickness of Feature C

1C) - Feature C

1. Feature C is a simply supported beam
2. The 1200-lbf load acts at the center of the beam
3. The bracket and loading are symmetric
4. Feature C has a rectangular cross-section
5. Direct shear stress is neglected

1D) - Feature C



1E) - Feature C

$$M_{max} = \frac{PL_c}{4} \quad \sigma = \frac{Mc}{I}$$

$$I = \frac{W_c h_c^3}{12} \quad c = \frac{h_c}{2}$$

$$\sigma_{allow} = \frac{6 M_{max}}{W_c h_c^2}$$

$$h_c = \sqrt{\frac{6 M_{max}}{W_c \sigma_{allow}}}$$

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4ES - Feature C

$$M_{max} = \frac{(1200)(2.4964)}{4} + M_{max} = 748.92 \text{ lb} \cdot \text{in}$$

$$h_c = \sqrt{\frac{6(748.92)}{(0.90)(9000)}} \quad h_c = \sqrt{0.6241}$$

$$h_c = 0.80 \text{ in}$$

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1A) - Feature D

Known Values:

$P = 1200 \text{ lbf}$, $SF = 4$, $S_y = 36,000 \text{ psi}$

$\sigma_{allow} = \frac{S_y}{SF} = \frac{36,000}{4} = 9000 \text{ psi}$

$b_D = 2.44 \text{ in}$ (width of Feature D)

$L_D = 1.50 \text{ in}$ (distance between supports)

1B) - Feature D

h_D = Minimum Required thickness of Feature D

1C) - Feature D

1. Feature D is modeled as a simply supported beam

2. The 1200-lbf load acts at the center of the beam

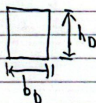
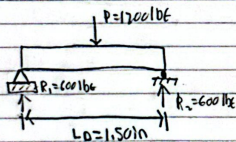
3. The loading and geometry are symmetric

4. Feature D has a rectangular cross-section with width b_D and thickness

h_D

5. Direct shear is neglected

1D) - Feature D



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4e) - Feature D

$$M_{max} = \frac{P L_0}{4} \Rightarrow \sigma = \frac{M_c}{I}$$

$$I = \frac{b h_D^3}{12} \Rightarrow C = \frac{h_D}{12} \Rightarrow \sigma_{allow} = \frac{6 M_{max}}{b h_D^2}$$

$$h_D = \sqrt{\frac{6 M_{max}}{b \sigma_{allow}}}$$

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1A) - Feature E

Known Values:

$$P_E = 600 \text{ lbf}, SF = 4, S_y = 36,000 \text{ psi}, \sigma_{allow} = 9000 \text{ psi}$$

$$b_E = 0.80 \text{ in}, L_E = 0.416 \text{ in}$$

1B) - Feature E

h_E = Minimum required thickness of Feature E

1C) - Feature E

1. Feature E is modeled as a simply supported beam

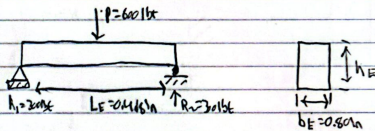
2. The 600-lbf load acts at the center of the beam

3. The loading and geometry are symmetric

4. Feature E has a rectangular cross-section with width b_E and thickness h_E

5. Direct shear is neglected

1D) - Feature E



1e) Feature E

$$M_{max} = \frac{P_E L_E}{4}, \sigma = \frac{M_c}{I}$$

$$I = \frac{b_E h_E^3}{12}, c = h_E / 2$$

$$\sigma_{allow} = \frac{6 M_{max}}{b_E h_E^2}$$

$$h_E = \sqrt[3]{\frac{6 M_{max}}{b_E \sigma_{allow}}}$$

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4A) - Feature E

$$M_{\max} = \frac{(600)(0.498)}{4} \quad , \quad M_{\max} = 74.715 \text{ in}$$

$$h_E = \sqrt{\frac{6(74.7)}{(0.8)(4000)}}$$

$$h_E = 0.66225$$

$$h_E = 0.25 \text{ in}$$

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2A) - Feature A

Known Values:

$P = 27 = 1200 \text{ lb}$, $E = 29,000,000 \text{ psi}$, $\delta_{allow} = 0.005 \text{ in}$, $L_A = 0.375 \text{ in}$

2B) - Feature A

r_A = minimum require radius

$d_A = 2r_A$

2C) - Feature A

1. Feature A is a cantilever beam

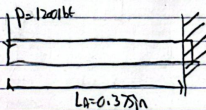
2. Resultant shop load acts 0.375 in from the fixed end

3. ASTM A36 steel is linearly elastic

4. Feature A is a solid circular section

5. Shear deflection is negligible

2D) - Feature A



2E) - Feature A

$\delta = \frac{PL^3}{3EI}$, $I = \frac{\pi d^4}{64}$

$\delta = \frac{4PL^3}{3E\pi d^4}$

$r_A = \left(\frac{4PL^3}{3E\pi \delta} \right)^{1/4}$

2F) - Feature A

$r_A = \left[\frac{4(1200)(0.375)^3}{3(29,000,000)\pi(0.005)} \right]^{1/4}$

$r_A = 0.117 \text{ in}$

$d_A = 0.233 \text{ in}$

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2A) - Feature B

$P = 1200 \text{ lbf}$, $E = 29,000,000 \text{ psi}$, $\epsilon_{allow} = 0.005 \text{ in}$, $L_B = 0.80 \text{ in}$, $b_B = 0.80 \text{ in}$

2B) - Feature B

$\epsilon_B = \text{Minimum required thickness}$, $A_B = b_B \epsilon_B$

2C) - Feature B

1. Feature B is axially loaded

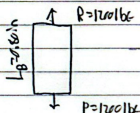
2. Load passes through the center

3. B has a rectangular cross-section

4. Axial length is taken as 0.80 in

5. Shear deformation is negligible

2d) - Feature B



2e) - Feature B

$$\epsilon = \frac{P L_B}{A_B E} \rightarrow A_B \epsilon_B \rightarrow \epsilon = \frac{P L_B}{b_B \epsilon_B E}$$

$$\epsilon_B = \frac{P L_B}{E b_B \epsilon_B}$$

2f)

$$\epsilon_B = \frac{(1200)(0.80)}{(29,000,000)(0.005)(0.80)} \rightarrow \boxed{\epsilon_B = 0.00826 \text{ in}}$$

$$A_B = b_B \epsilon_B \rightarrow A_B = (0.8)(0.00826)$$

$$\boxed{A_B = 0.006608 \text{ in}^2}$$

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2A) - Feature C

$P = 1200 \text{ lb}$, $E = 29,000,000 \text{ psi}$, $G_{\text{steel}} = 0.005 \text{ in}$

$L_c = a + 2b = 2.4464 \text{ in}$, $b_c = 0.60 \text{ in}$

2B) - Feature C

h_c = minimum required thickness

2C) - Feature C

1. Feature C is a simply supported beam

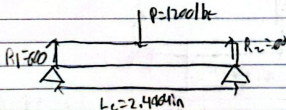
2. The 1200-lb load acts at midspan

3. Geometry and loading are symmetric

4. C has a rectangular cross-section

5. Shear deflection is negligible

2D) - Feature C



2E) - Feature C

$\delta = \frac{PL_c^3}{48EI}$, $F = \frac{bch_c^3}{12}$, $\delta = \frac{PL_c^3}{4Ebh_c^3}$

$h_c = \left(\frac{PL_c^3}{4Ebg} \right)^{1/3}$

2F)

$h_c = \left[\frac{(1200)(2.4464)^3}{4(29 \times 10^6)(0.60)(0.005)} \right]^{1/3}$

$h_c = 0.343 \text{ in}$

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2A) - Feature D

$P_D = 600 \text{ lb}$, $E = 29,000,000 \text{ psi}$, $\delta_{allow} = 0.005 \text{ in}$, $L = 1.44 \text{ in}$, $b_D = 0.8 \text{ in}$

2b) - Feature D

$t_D = \text{minimum required thickness}$

$A_D = b_D t_D$

2C) - Feature D

1. Each side carries $\frac{1}{2} P_D$ by symmetry

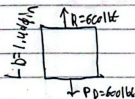
2. Feature D is an axial side member

3. Load-transfer Length is $L = 1.44 \text{ in}$

4. D has a rectangular cross-section

5. Shear deformation is negligible

2D) - Favored



2e) - Feature D

$$\delta = \frac{P_D L}{A_D E} \quad , \quad A_D = b_D t_D \quad , \quad \delta = \frac{P_D L}{b_D t_D E}$$

$$t_D = \frac{P_D L}{E \delta b_D}$$

2E) - Feature D

$$t_D = \frac{(600)(1.44 \text{ in})}{(29 \times 10^6)(0.005 \text{ in})(0.8 \text{ in})} \rightarrow t_D = 0.0078 \text{ in}$$

$$A_D = 0.00624 \text{ in}^2$$

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2A5- Feature E

$P_E = 600 \text{ lb}$, $E = 29,000,000 \text{ psi}$, $\delta_{allow} = 0.005 \text{ in}$, $L_E = b = 0.9992 \text{ in}$, $b_E = 0.8 \text{ in}$

2B- Feature E

$h_E = \text{Minimum required thickness}$

2C- Feature E

1. Feature E is treated as a cantilever lip

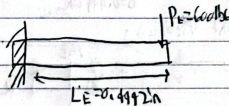
2. One side carries the 600-lb reaction

3. Lip length is $b = 0.9992 \text{ in}$

4. E has a rectangular cross-section

5. Shear deflection is negligible

2d)- Feature E



2e)- Feature E

$$\delta = \frac{P_E L_E^3}{3EI}, \quad I = \frac{b_E h_E^3}{12} \Rightarrow \delta = \frac{4P_E L_E^3}{E b_E h_E^3}$$

$$h_E = \left(\frac{4P_E L_E^3}{E b_E \delta} \right)^{1/3}$$

2f)- Feature E

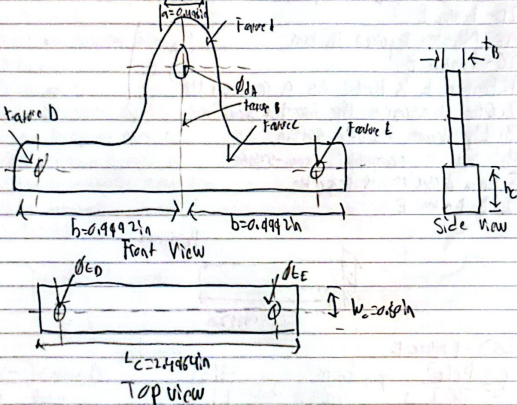
$$h_E = \left[\frac{4(600)(0.9992)^3}{(29 \times 10^6)(0.8)(0.005)} \right]^{1/3}$$

$$h_E = 0.274 \text{ in}$$

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3.

Multiview Sketch - Stress Analysis



Notes (Stress Analysis):

1. Overall geometry based on given dimensions from Assignment

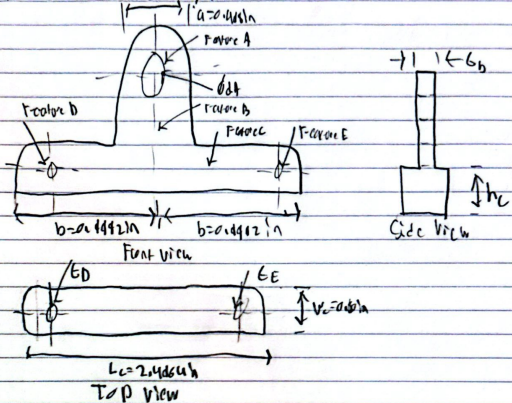
2. Features:

- A: pin hole (diameter d_a)
- B: vertical web (thickness b)
- C: horizontal base (thickness h , width w_c , length L_c)
- D: left mounting hole (diameter d_b)
- E: right mounting hole (diameter d_c)

3. Dimensions shown are used for Stress Analysis in Part 1

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Multiview Sketch - Stiffness Analysis



Notes (Stiffness Analysis)

1. Same overall geometry as stress analysis
2. Deflection of 0.005 in is used for each feature
3. Feature analyzer for stiffness:

- A: pin hole (diameter d_A)
- B: Vertical web (thickness t)
- C: horizontal base (thickness h_c , width w_c , length L_c)
- D: Left mounting hole (diameter d_D)
- E: Right mounting hole (diameter d_E)

4. Dimensions shown are used for stiffness analysis in part 2